

KOTA BHARU, Malaysia

WMO: 486150

Lat: 6.1638N

Lon: 102.3010E

Elev: 5

StdP: 101.27

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.1	22.8	20.0	14.7	25.0	20.8	15.5	25.5	8.5	27.3	8.0	27.4	1.1	210	0.339

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.2	33.2	26.8	32.9	26.7	32.2	26.5	27.9	31.7	27.6	31.3	27.2	31.0	3.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	30.6	26.4	21.8	30.0	26.1	21.6	29.8	89.7	31.4	87.6	31.9	86.3	31.3	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.2	5.4	20.9	34.6	1.0	0.7	20.2	35.2	19.6	35.6	19.0	36.0	18.3	36.6		
(5)			20.0	28.9	1.3	0.7	19.1	29.4	18.3	29.7	17.6	30.1	16.7	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	26.4	26.8	27.6	28.6	28.8	28.5	28.1	28.0	27.8	27.4	26.7	26.5		
	DBStd	1.18	0.98	0.87	0.98	0.90	0.86	0.71	0.79	0.75	0.62	0.85	1.04	1.01		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6422	509	471	545	557	582	554	562	557	533	539	502	511		
	CDD18.3	3380	251	238	287	307	324	304	303	298	283	281	252	253		
	CDH23.3	34953	2257	2297	3042	3634	3823	3421	3264	3116	2853	2740	2252	2254		
	CDH26.7	12168	459	608	1046	1511	1642	1413	1276	1198	1052	889	588	486		
(14) Wind	WSAvg	2.3	3.0	3.0	2.6	2.3	2.1	1.9	1.9	2.0	1.9	1.9	1.9	2.7		
(15) Precipitation	PrecAvg	2662	119	53	117	96	105	123	160	166	178	272	668	569		
	PrecMax	3717	620	272	662	379	283	295	346	347	393	596	1614	1543		
	PrecMin	1540	0	0	1	0	0	0	45	47	58	80	166	43		
	PrecStd	548	127	57	146	89	60	62	71	70	76	116	346	318		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	31.3	32.8	34.1	34.2	33.2	33.2	33.2	32.8	32.2	31.3	31.0		
		MCWB	25.9	25.6	26.3	26.8	27.1	26.9	26.6	26.0	25.9	26.2	26.2	26.5		
	2%	DB	30.0	30.8	32.0	33.1	33.2	32.9	32.3	32.2	32.0	31.6	30.8	30.1		
		MCWB	25.5	25.5	26.1	26.7	27.1	26.8	26.5	26.1	26.1	26.2	26.3	25.9		
	5%	DB	29.2	30.1	31.2	32.6	32.8	32.2	32.0	31.9	31.4	31.0	30.2	29.6		
		MCWB	25.0	25.2	26.0	26.6	27.0	26.6	26.4	26.2	26.1	26.2	26.1	25.5		
	10%	DB	28.9	29.7	30.8	32.0	32.1	31.9	31.3	31.1	31.0	30.4	29.7	29.0		
		MCWB	24.8	25.0	25.9	26.5	26.7	26.6	26.3	26.1	26.1	26.1	25.9	25.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.6	27.4	28.1	28.4	28.5	28.0	27.5	27.6	27.2	27.2	27.0		
		MCDB	29.4	30.3	30.9	32.3	32.4	31.9	31.4	30.9	30.9	30.7	30.0	29.8		
	2%	WB	26.1	26.1	26.7	27.5	27.8	27.7	27.4	27.0	27.0	26.8	26.7	26.4		
		MCDB	28.9	29.7	30.6	31.7	31.8	31.3	31.0	30.7	30.7	30.3	29.6	29.0		
	5%	WB	25.7	25.7	26.5	27.1	27.5	27.3	27.0	26.7	26.6	26.5	26.3	26.0		
		MCDB	28.4	29.1	30.2	31.2	31.5	31.0	30.7	30.3	30.3	29.9	29.1	28.6		
	10%	WB	25.3	25.4	26.1	26.7	27.1	26.9	26.6	26.4	26.3	26.2	26.0	25.7		
		MCDB	27.9	28.6	29.7	30.6	31.0	30.7	30.2	30.0	29.9	29.5	28.8	28.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	6.0	6.6	7.1	7.2	7.3	7.2	7.3	7.2	6.3	5.2	4.8		
		MCDBR	5.6	6.7	7.2	7.7	7.9	7.7	7.7	7.8	7.6	7.1	6.3	5.6		
		MCWBR	2.4	2.7	2.8	2.9	3.3	3.5	3.4	3.3	3.4	3.1	2.8	2.5		
	5% WB	MCDBR	5.1	5.9	6.7	7.2	7.4	7.3	7.3	7.2	7.2	6.7	5.8	5.1		
		MCWBR	2.5	2.7	3.0	3.1	3.5	3.7	3.6	3.5	3.5	3.2	2.8	2.7		
(40) Clear-Sky Solar Irradiance	taub		0.464	0.456	0.461	0.445	0.454	0.471	0.476	0.474	0.471	0.466	0.450	0.473		
	taud		2.301	2.306	2.309	2.366	2.350	2.321	2.299	2.290	2.295	2.319	2.365	2.285		
	Ebn at Noon		856	873	867	867	840	815	814	831	846	853	862	839		
	Edh at Noon		134	136	136	127	125	127	131	135	136	133	124	134		
(44) All-Sky Solar Radiation	RadAvg		4.50	5.54	5.84	5.90	5.33	5.13	5.08	5.28	5.24	4.63	3.97	3.81		
	RadStd		0.36	0.40	0.51	0.29	0.28	0.20	0.29	0.16	0.23	0.31	0.30	0.43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.34	+0.23	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.16	N/A	N/A	+0.37	+0.28	+0.21	N/A	N/A	N/A	N/A

Nomenclature: See separate page